NET731 Week10 Azure Environment Architecture

Slide 1 — Title Slide

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Slide 2 — Project Overview

Overview of the Project

Environment Created

- Built a cloud networking setup in Azure
- Deployed and configured virtual machines
- Set up virtual networks and subnet configurations
- Applied security settings and controls
- Implemented monitoring and cost tracking tools

Main Objectives

- Develop practical skills in Azure cloud management
- Build and manage secure cloud infrastructure
- Track and optimise resource usage and performance
- Gain hands-on experience with Azure services and tools

Slide 3 — Architecture Diagram

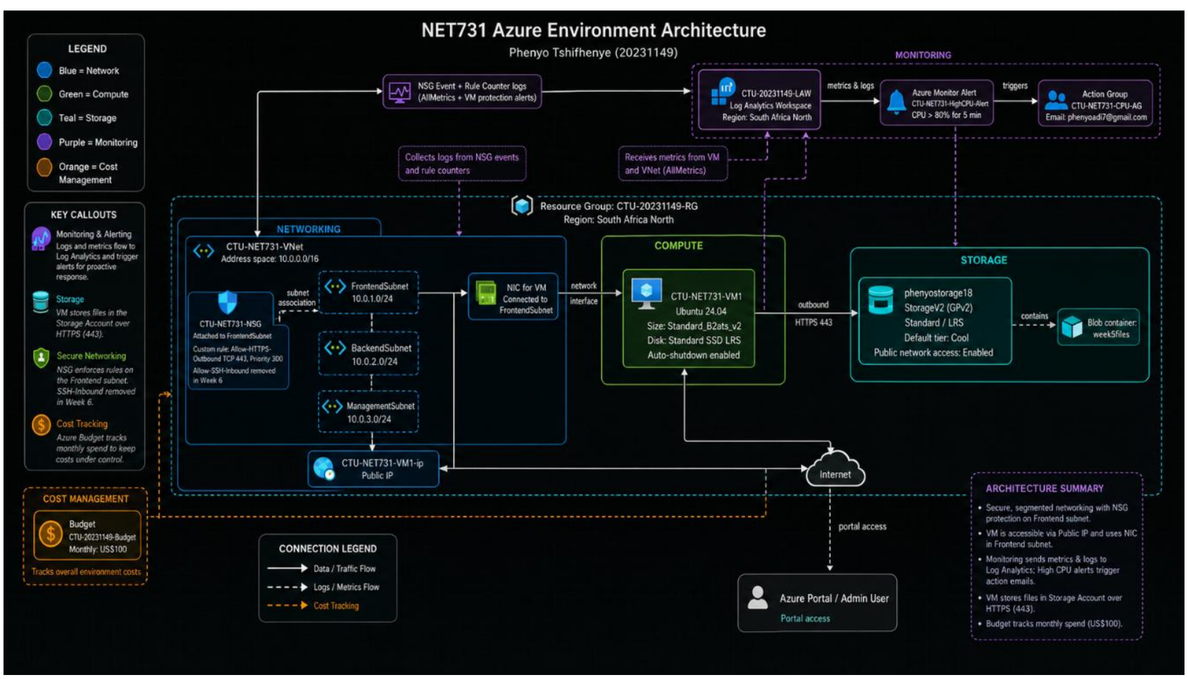
Azure Architecture Layout

Components Used

- Resource Groups
- Virtual Network (VNet)
- Subnets
- Network Security Groups (NSGs)
- Virtual Machine (VM)
- Storage Account
- Azure Monitor
- Log Analytics Workspace

Description

The diagram illustrates how the networking, security, monitoring, and storage services interact and connect within the Azure cloud environment.



Slide 4 — Key Technical Decisions

Important Technical Choices

Decision 1 — VM Selection

- Chose the B1s virtual machine size
- More affordable option for the project
- Appropriate for a testing and learning environment

Decision 2 — NSG Setup

- Configured inbound Remote Desktop Protocol (RDP) access
- Limited unnecessary access where possible
- Strengthened overall network protection

Decision 3 — Monitoring Configuration

- Activated Azure Monitor
- Set up a Log Analytics Workspace
- Enabled performance and resource monitoring

Slide 5 — Security and Monitoring

Security and Monitoring Features

Security Measures Applied

- Configured Network Security Groups (NSGs)
- Implemented RBAC access permissions
- Used Microsoft Defender for Cloud recommendations
- Applied access management settings

Monitoring Features Used

- Created Azure Monitor alerts
- Monitored CPU and network activity
- Configured budget notifications
- Used Log Analytics Workspace for tracking logs

Importance

These tools assisted in improving security, monitoring system performance, and identifying potential performance or cost-related problems early.

Slide 6 — Cost Management

Managing Cloud Costs

Cost Reduction Methods

- Configured Azure budget alerts
- Tracked expenses using Cost Analysis
- Shut down unused virtual machines
- Selected lower-cost VM options

Estimated Spending

- Expected costs stayed under $100
- The virtual machine generated the highest costs

Key Lesson

Cloud services can become costly if resources are not monitored and managed carefully.

Slide 7 — Challenges and Solutions

Problems Experienced and Solutions Applied

Challenge 1

Some VM sizes could not be deployed

Solution

Changed the Azure region and selected different VM sizes

Challenge 2

Network connectivity and NSG configuration problems

Solution

Checked inbound NSG rules and fixed the RDP configuration

Challenge 3

Difficulty controlling Azure expenses

Solution

Stopped and deallocated resources that were not in use

Slide 8 — Lessons Learned

Knowledge and Skills Gained

- Azure networking configuration
- Virtual machine deployment
- Cloud monitoring techniques
- Security management
- Cost management and optimisation
- GitHub documentation skills

Improvements for Future Projects

- Plan resource deployment earlier
- Keep screenshots organised throughout the project
- Make greater use of automation tools

Slide 9 — Conclusion

Summary

Main Outcomes

- Successfully created an Azure cloud environment
- Configured networking, monitoring, and security settings
- Controlled and monitored cloud costs effectively
- Improved practical knowledge of cloud administration

Final Thoughts

This project offered useful hands-on experience with Microsoft Azure and modern cloud networking technologies.